

Technical Laboratory Report: **The Light-Sensitive Switch (LDR+ Voltage Divider)**

Connector Dilara Rzazada, Nazrin Aliyeva, Ruslan Qasimov,
Ulker Aliyeva, Khanverdi Chelebiyev, Nazila Aliyeva, Sabina
Qarakshiyeva, Imran Cherchiyev

Verified by: Physics Teacher Azerbaijan Telman Askeraliyev
(Fizika muellimi, Azerbaijan, Baku)

Project Documentation: The Light-Sensitive Switch (LDR+ Voltage Divider)

Simulating a Voltage Divider Circuit in Proteus

This documentation outlines the steps to design and simulate a basic voltage divider circuit using Proteus design software.

1. Objective

To demonstrate the voltage divider rule, which allows for a specific output voltage to be derived from a higher input voltage by using two resistors in series.

2. Components Required

DC Power Supply: Set to 10V to provide the input voltage.

Resistors: Two 10-ohm, 1.6-watt metal film resistors.

DC Voltmeter: Used to measure the output voltage in the middle of the circuit .

Ground: A necessary terminal added to the circuit to provide a reference point for voltage measurements

3. Circuit Configuration

Placement: Place the DC supply and two resistors in a workspace. Rotate resistors clockwise if needed for clean layout .

Wiring: Connect the components in series, starting from the positive terminal of the DC supply, through the first resistor, the second resistor, and finally to the ground .

Measurement: Place the DC voltmeter across the second resistor (R2) and the ground to measure the output voltage .

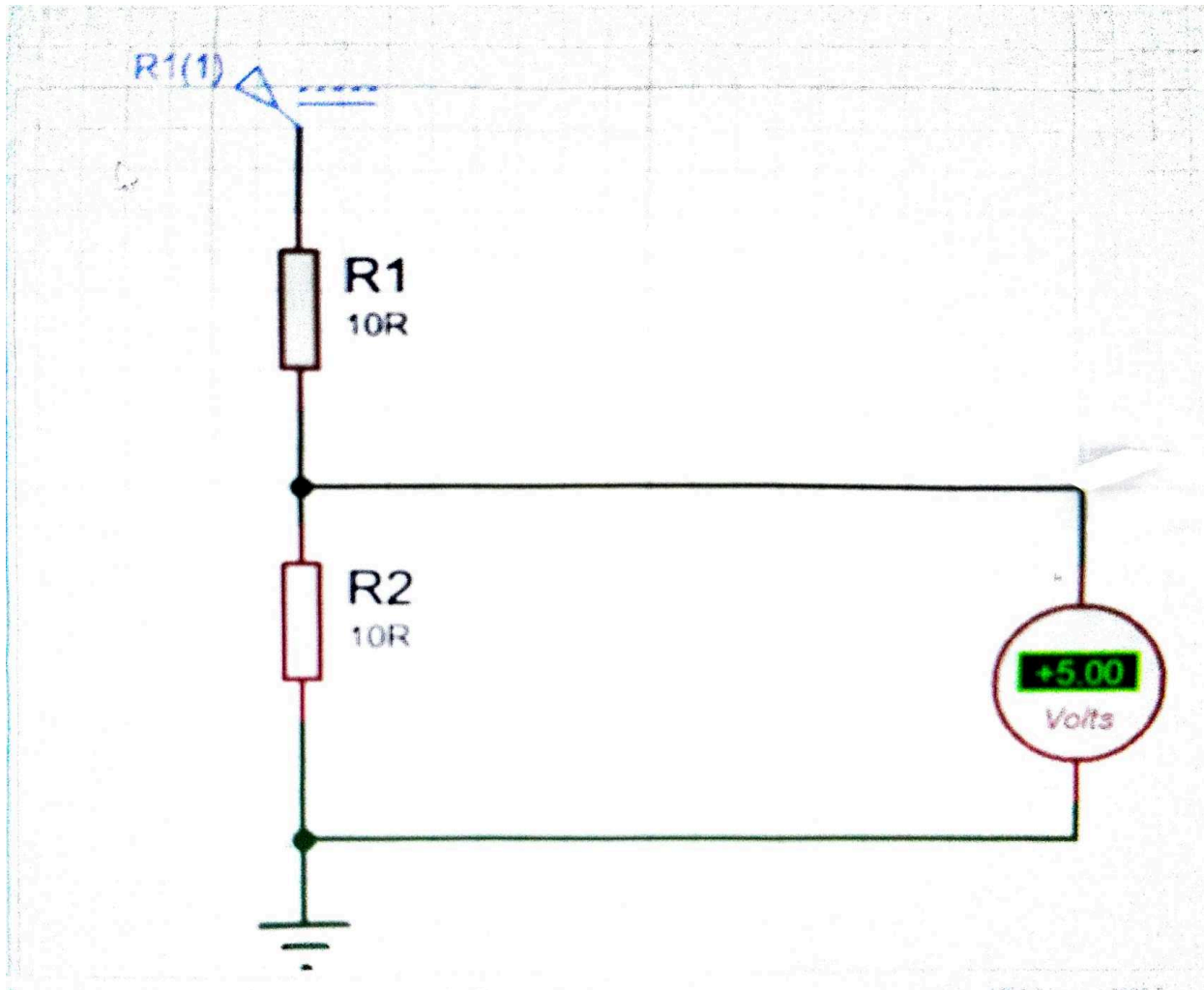
4. Mathematical Formula

The voltage divider rule used to calculate the output voltage (V_out) is:

$$V_{out} = V_{in} * (R2 / (R1 + R2))$$

Using the values from the video (V_in = 5V, R1 = 10 ohm, R2 = 10 ohm):

$$V_{out} = 5 * (10 / (10 + 10)) = 5 * (10 / 20) = 2.5 \text{ Volts .}$$



5. Simulation Results

Upon running the simulation, the voltmeter connected across R2 should display 2.5V. The remaining 2.5V is dropped across the first resistor (R1), confirming that the total supply voltage is divided equally when R1 and R2 are identical.

